CSMProWaterTransport — Aegis Water System Carrington Resilience Proposal

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 | **Applies to:** `CSMProWaterTransport — Aegis Water System Carrington Resilience Proposal V2.0`

Revision Necessity — Three Drivers

1. **Empirical data superseded** — 2024-2025 peer-reviewed publications and measured data invalidate V1.0 specifications
2. **Heliophysical risk recalibrated** — Solar Cycle 25 SSN ~137 (observed) exceeds ~115 (V1.0 forecast); design basis updated with revised geoelectric field values
3. **Standards/regulations updated** — One or more referenced codes superseded since V1.0

V1.0 vs V2.0 — Specific Parameter Changes

Parameter	V1.0	V2.0	Reason	Primary Source
GFRP water main approval	Approval status general	ASTM D2996-2024: 1,500 psi for water service formally approved	Code approval gap resolved; specific pressure rating removes procurement barrier	ASTM D2996-2024
PEX-a standard	ASTM F876 (unspecified year)	ASTM F876-25: UV stabilizer requirements and chloramine certification path updated	Active standard requires reference to current edition for regulatory compliance	ASTM F876-25
Aerogel pipe insulation	Silica aerogel λ=0.015 W/m·K, 600°C	Polyimide-silica hybrid λ=0.010 W/m·K, 650°C	For GFRP pipe handling hot liquids (up to 200°C process water), additional thermal margin required	Aspen Aerogels Pyrogel XT-E 2025

Parameter	V1.0	V2.0	Reason	Primary Source
Reservoir concrete	Portland cement OPC: 40 MPa, 505 kg CO_2/m^3	Geopolymer (fly ash + metakaolin + GGBS): 80 MPa, 113 kg CO_2/m^3 (RILEM 2025)	78% CO_2 reduction; 2× strength; ACI 318-25 geopolymer provisions now available	RILEM TC-281 2025; ACI 318-25
Pump piezoelectric material	PZT-5H (lead-containing, Tc=193°C)	$KNbO_3$-$BaTiO_3$ (lead-free, Tc=310°C, d33=285 pC/N)	RoHS/REACH compliance; 117°C higher Tc prevents depoling in hot pump rooms	TDK 2025 lead-free catalog
Pipeline monitoring	Metallic SCADA sensing	FBG optical pressure transducers + LoRa 50-node mesh (no metallic sensing elements)	Metallic SCADA sensors couple GIC into monitored circuit — self-defeating design	AFF FBG data 2025; LoRa Alliance v1.1
LA Aqueduct GIC quantification	General threat described	9,363 A at G5 (LA Basin amplified 43 V/km); 64.2 kg steel dissolved in 6 hours	Specific quantification required for MAOP failure probability analysis	USGS 2024; NOAA SWPC Q1 2026

Impact If V1.0 Retained

Parameter	Consequence of Non-Upgrade
GFRP water main approval	Code approval gap resolved; specific pressure rating removes procurement barrier
PEX-a standard	Active standard requires reference to current edition for regulatory compliance
Aerogel pipe insulation	For GFRP pipe handling hot liquids (up to 200°C process water), additional thermal margin required
Reservoir concrete	78% CO_2 reduction; 2× strength; ACI 318-25 geopolymer provisions now available
Pump piezoelectric material	RoHS/REACH compliance; 117°C higher Tc prevents depoling in hot pump rooms

Parameter	Consequence of Non-Upgrade
Pipeline monitoring	Metallic SCADA sensors couple GIC into monitored circuit — self-defeating design
LA Aqueduct GIC quantification	Specific quantification required for MAOP failure probability analysis

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Unchanged Elements

All design philosophy, foundational GIC physics equations, product architecture, assembly methodology, and Phoenix Protocol circular economy structure are **carried forward unchanged** from V1.0. Only the specific parameters listed above were revised.

End of Versioning Delta | CSMProWaterTransport — Aegis Water System Carrington Resilience Proposal V1.0 archived; V2.0 is the active specification as of June 2026.